

FEHMI OZBAYRAK

 [linkedin.com/in/fozbayrak](https://www.linkedin.com/in/fozbayrak)

 github.com/fozba

 ozbayrakfehmi@gmail.com

 1-512-920-8349

SUMMARY

PhD Candidate in Petroleum Engineering with dual expertise in **numerical simulation** (geomechanics, reservoir modeling) and **applied machine learning**. Seeking an internship to contribute to complex geothermal reservoir characterization and performance optimization challenges.

EDUCATION

The University of Texas at Austin

Austin, TX

PhD Candidate in Petroleum Engineering (Exp. Grad: 2027) | GPA: 3.67/4.00

2022 – Present

- **Research:** Computational Geomechanics & Data-Driven Subsurface Modeling (Adv: Dr. Mukul M. Sharma)
- Completed Master of Science in Petroleum Engineering en route (2024) in data analytics and machine learning in geosciences (Adv: Dr. Michael J. Pyrcz & Dr. John T. Foster).

Middle East Technical University

Ankara, Turkey

Bachelor of Engineering in Petroleum Engineering | GPA: 3.21/4.00

2018 – 2022

SKILLS

Reservoir Engineering	Geothermal Energy, Geomechanics, Reservoir Simulation, History Matching, Fiber Optic Data Analysis (DAS/DTS), Geostatistics
Modeling	Numerical Simulation (FEM/FVM), Techno-Economic Analysis, ML, Optimization
Software & Tools	ResFrac, OpenFOAM, Python (Pandas, PyTorch, Scikit-learn), Linux, VBA, C/C++ , PowerBI

EXPERIENCE

Graduate Research Assistant | Center for Integrated Seismicity Research (CISR)

2024 – Present

- Developing comprehensive, coupled-process geomechanical models of the Delaware Mountain Group to investigate the risks of wastewater injection-induced fracturing and seismicity.
- Building and simulating models using advanced fractured reservoir simulators (ResFrac, OpenFOAM) by integrating diverse datasets, including well logs, public injection data, and regional structural models.
- Validating model accuracy by history-matching simulation results against real-world bottom-hole pressure and temperature data from active saltwater disposal (SWD) wells.
- Leveraging validated models to conduct sensitivity studies and investigate safe “injection envelopes,” providing insights into fracture containment and propagation under various operational scenarios.

Reservoir Engineering Intern | Fervo Energy – Houston, TX

May 2025 – Aug 2025

- Architected and developed a modular, object-oriented Python platform for the rapid techno-economic forecasting of geothermal assets, integrating ML proxy models with a sophisticated physics engine.
- Achieved a >99% reduction in simulation runtime (from hours to seconds) by replacing direct reservoir simulation with a dual-timescale engine powered by ML proxy models.
- Integrated a Bayesian Optimization engine to automate the search for revenue-maximizing drilling schedules, transforming the platform from a predictive to a prescriptive strategic tool.
- Developed an interactive Dash-based GUI, empowering engineers and analysts to intuitively build, run, and visualize complex multi-facility and multi-contract development scenarios.
- Established a professional software foundation with comprehensive, enterprise-grade documentation, a reproducible environment, and a documented API to ensure long-term maintainability.

Graduate Research Assistant | DiReCT Consortium

2022 – 2024

- Conducted research in spatial statistics and machine learning, developing a published algorithm to improve uncertainty quantification in subsurface modeling.

Field Engineer Intern | Middle East Air Drilling Services – Alasehir, Turkey

2021

- Provided critical operational support on a geothermal drilling rig, taking the lead in documenting a cement job failure; the detailed report was used as the primary reference in the investigation.
- Gained end-to-end field experience, from rig-up to casing, and actively observed key technologies including MWD, coiled-tubing, and cementing.

SELECTED PROJECTS

- Molecule-to-Megabyte Gas-to-Data-Center Techno-Economic Model** | *SPE E3S Energy Datathon (2nd Place Winner)* 2026
- Built a full techno-economic digital twin for gas-powered AI data centers, integrating physics-based generation modeling with project finance
 - Presented a bankable on-site power strategy using modular natural gas engines, stochastic risk analysis, and IRR/DSCR optimization to industry judges.
- Utah FORGE EGS Fracture Network Model** | *GEODE Datathon (2nd Place Winner)* 2025
- Analyzed the Utah FORGE EGS dataset by integrating multi-source data (tracers, fiber optics, microseismic) to build a holistic model of the fracture network.
 - Diagnosed that dominant flow paths were creating a ‘stress shadowing’ phenomenon, risking premature thermal short-circuiting and leaving significant reservoir volume under-utilized.
 - Proposed a targeted re-stimulation strategy to enhance flow uniformity and extend the project’s thermal lifetime.
- Nevada Play Fairway CNN Explorer** | *GEODE Datathon (2nd Place Winner)* 2025
- Developed a novel CNN-based image processing workflow that significantly improves the predictive power of the Nevada Play Fairway Analysis for geothermal exploration.
- Hydraulic Fracturing Energy Forecaster** | *UT PGE Hackathon (1st Place Winner)* 2025
- Developed a Python-based ML workflow to forecast hydraulic fracturing energy consumption (Grid, Diesel, CNG) and optimize supply chain logistics.
- Discrete Fracture Network Characterization** | *SPE Geothermal Datathon (2nd Place Winner)* 2023
- Analyzed the Utah FORGE EGS microseismic event catalogue to predict the discrete fault network using tree-based ensemble machine learning methods.

LEADERSHIP & SERVICE

- President** | *Switch Energy Alliance (UT Chapter)* 2026 – Present
- President of the UT Chapter; leading a student organization focused on promoting energy literacy through campus-wide events and industry engagement.
- Hackathon Architect** | *UT PGE Energy AI Hackathon* 2024
- Co-architected the 2024 Hackathon, designing the technical challenge, curating datasets, and mentoring teams.